SESSION 5
DISTINCT, ORDER BY & LIMIT

SQL Answers with Full Questions and Easy Explanations

Topics: DISTINCT • Sorting (ASC, DESC) • Limiting rows

Task 1

Question: Write an SQL query using the DISTINCT keyword to find all unique payment methods used in the orders table of a food delivery app database.

SQL Answer

```
SELECT DISTINCT payment_method
FROM orders;
```

Easy Explanation

This query displays each payment method used in the orders table only once.

- SELECT payment_method chooses the payment_method column to display.
- DISTINCT removes repeated values from the result.
- FROM orders tells SQL to read the data from the orders table.

Task 2

Question: Query the users table to list all cities where users have registered, but display each city only once and sort the result in alphabetical order (A-Z).

SQL Answer

```
SELECT DISTINCT city
FROM users
ORDER BY city ASC;
```

Easy Explanation

This query lists unique cities and arranges them alphabetically from A to Z.

- DISTINCT city removes duplicate city names.
- ORDER BY city sorts the result by city name.
- ASC means ascending order, which is A to Z for text.

Task 3

Question: Write an SQL query to select the top 5 most recent movie bookings from the bookings table, ordered by booking_date in descending order.

SQL Answer

```
SELECT *
FROM bookings
ORDER BY booking_date DESC
LIMIT 5;
```

Easy Explanation

This query returns the five most recent bookings by placing the newest booking dates first.

- SELECT * displays all columns from each selected booking.
- ORDER BY booking_date DESC sorts booking dates from newest to oldest.
- LIMIT 5 returns only the first five rows after sorting.

Task 4

Question: From a products table containing Flipkart-style product data (id, name, category, sold_count), write an SQL query to retrieve the 10 products with the highest sold_count, displaying only product name and sold_count, sorted from highest to lowest. Use ORDER BY and LIMIT together.

SQL Answer

```
SELECT name, sold_count
FROM products
ORDER BY sold_count DESC
LIMIT 10;
```

Easy Explanation

This query displays the ten products with the largest sold_count values, from highest to lowest.

- SELECT name, sold_count displays only the two requested columns.
- FROM products reads product records from the products table.
- ORDER BY sold_count DESC puts the most-sold products first.
- LIMIT 10 restricts the result to ten products.

Quick Revision

- DISTINCT: Returns unique values only, for example SELECT DISTINCT city FROM users;
- ORDER BY city ASC: Sorts text alphabetically from A to Z.
- ORDER BY sold_count DESC: Sorts numbers from highest to lowest.
- LIMIT 10: Restricts the result to ten rows.